

# JUNWEN LIU

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## APPOINTMENTS AND EDUCATION

**The University of Hong Kong**, Hong Kong

Aug. 2026 – Present

*Postdoctoral Fellow*

Supervisor: Prof. Zhigang Bao

**Fudan University**, Shanghai, China

Sep. 2021 – Jun. 2026

*Ph.D. in Mathematics*

Supervisor: Prof. Lun Zhang

**Nankai University**, Tianjin, China

Sep. 2017 – Jul. 2021

*B.Sc. in Mathematics and Applied Mathematics*

## RESEARCH INTERESTS

Random matrix theory; asymptotic analysis; Painlevé equations; universality

## PUBLICATIONS

**Asymptotics for the noncommutative Painlevé II equation**

Junwen Liu, Luming Yao and Lun Zhang

*Nonlinearity* 39 (2026), 015017.

**Large gap asymptotics of the hard edge tacnode process**

Junwen Liu, Luming Yao and Lun Zhang

*J. Lond. Math. Soc.* 112 (2025), e70314.

## HONORS AND AWARDS

**Best Poster Award**, The 2026 Joint Workshop of RMTA & SNAB

2026

**National Scholarship (China)**

2025

## CONFERENCES

Invited Talks

**Workshop on Integrable Systems and Random Matrix Theory**

Great Bay University, Jan. 4–16, 2026

**Workshop on Integrable Systems and Random Matrix Theory**

Great Bay University, Jan. 5–17, 2025

Contributed Talks

**Fifth ZiF Summer School: Randomness in Physics and Mathematics**

ZiF, Aug. 25–Sep. 6, 2025

**AI for Math Graduate Academic Forum**

Shanghai Jiao Tong University, May 7–9, 2025

Other Conferences Attended

**Universality, Nonlinearity, and Integrability in Honor of Percy Deift**

KIAS, May 12–16, 2025

**Random Matrix Theory Summer School**

University of Michigan, Jun. 17–28, 2024

**Random Matrices and Related Topics in Jeju**

Lotte Hotel, Jeju Island, May 6–10, 2024

## TEACHING EXPERIENCE

**School of Mathematical Sciences, Fudan University**

Teaching Assistant, Basic Number Theory

Spring 2022

Teaching Assistant, Advanced Algebra I

Fall 2021

## SKILLS

- **Computer Skills:** Mathematica, LaTeX, Microsoft Office
- **Languages:** Chinese (Native), English (Fluent)